

Unit 4, Lesson 4: Distance on the Coordinate Plane



Benchmark

MA.6.GR.1.2 Find distances between ordered pairs, limited to the same x -coordinate or the same y -coordinate, represented on the coordinate plane.



Learning Target

- ☐ I can find the distance between two ordered pairs.



4.4.1 Warm-Up: A Point on the Number Line

1. Circle all numbers that could be the location of point F on the number line.

3.5

$4\frac{1}{2}$

$-\frac{3}{5}$

-3.5

$-\frac{7}{2}$

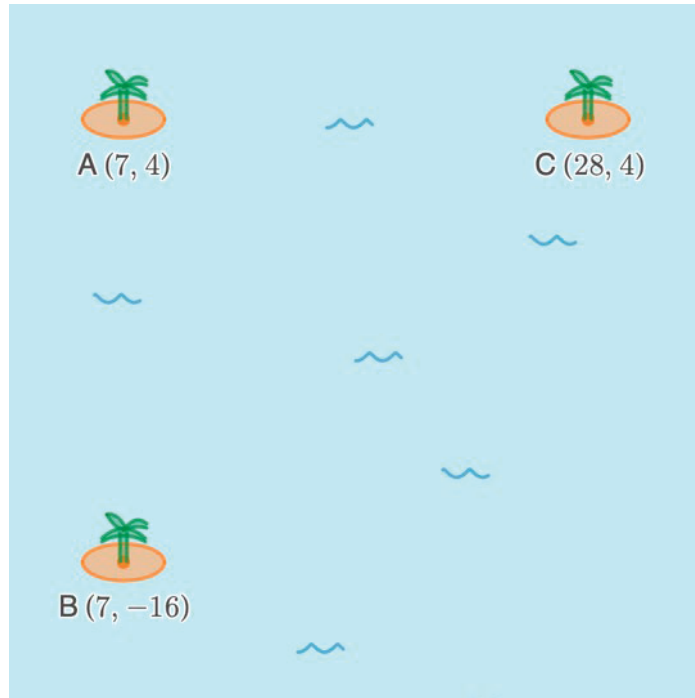
$-4\frac{1}{2}$





4.4.2 Collaboration: Who Is Correct?

The locations of three islands in the ocean are given using ordered pairs.



Three students described the distance between each island.

Braxton says the distance from point A to point C is **less than** the distance from point A to point B .

Justine disagrees with him and says the distance from point A to point C is **greater than** the distance from point A to point B .

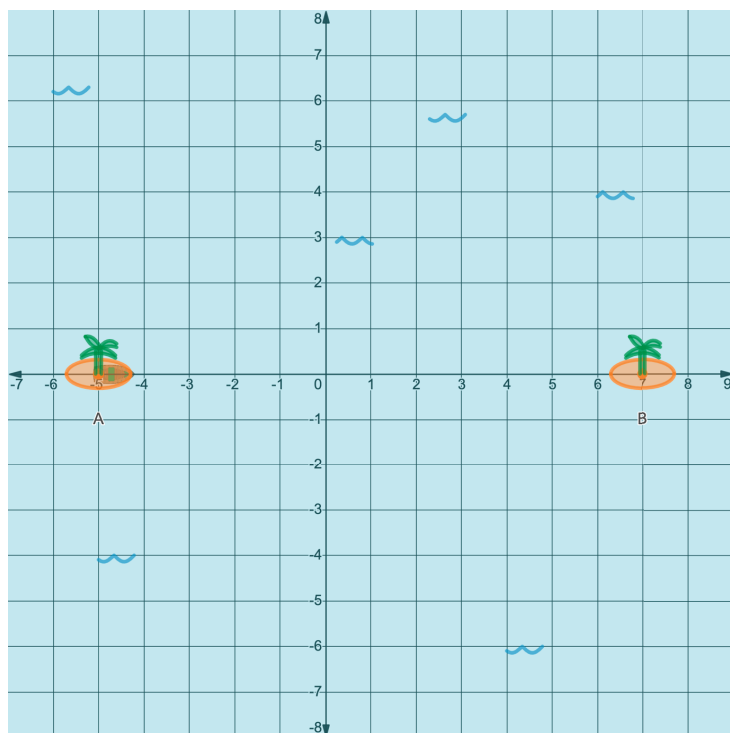
Rainey says the distance from point A to point C is **equal to** the distance from point A to point B .

1. Discuss with your partner who you think is correct. Write a summary of your discussion.



4.4.3 Guided Instruction: Distance on the Coordinate Plane

1. The coordinates of two small islands are labeled points A and B on the graph.



- a. Counting the grid lines, how far apart are the islands?
- b. What are the coordinates of points A and B ?
- A : (____, ____)
- B : (____, ____)

- c. Explain why the y -coordinates of the two points are the same.

- d. Describe the location of the two points relative to the y -axis.



The distance a number is from zero (0) on a number line is called the **absolute value**.

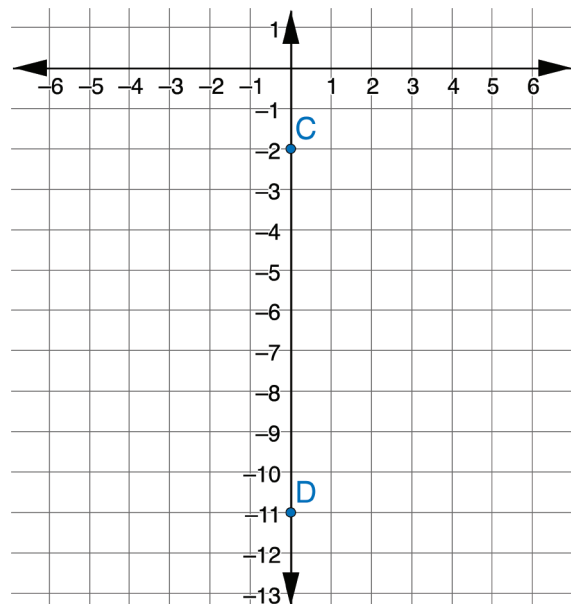
- e. What is the absolute value of the x -coordinate of each point?

The absolute value of the x -coordinate of Point A is ____.

The absolute value of the x -coordinate of Point B is ____.

- f. Write a mathematical sentence using absolute value that can be used to determine how far apart the two islands are.

2. Below is another coordinate plane with two different islands located at points C and D .



- How is the location of these two islands different from the first two you worked with?
- Counting the grid lines, how far apart are the islands?
- What are the coordinates of points C and D ?

C : (____, ____)

D : (____, ____)

- Explain why the x -coordinates of the two points are the same.
- Describe the location of the two points relative to the x -axis.
- What is the absolute value of the y -coordinate of each point?
The absolute value of Point C is _____.
The absolute value of Point D is _____.
- Write a mathematical sentence using absolute value that can be used to determine how far apart the two islands are.

3. Complete the statements.

When finding the horizontal distance between two

points that lie on the same line, if the points are on

opposite sides of the y -axis, ☐ add
☐ subtract the absolute
value of the unlike coordinates. If the points are on the

same side of the y -axis, then ☐ add
☐ subtract the absolute
values of the unlike coordinates.

When finding the vertical distance between two points

that lie on the same line, if the points are on the

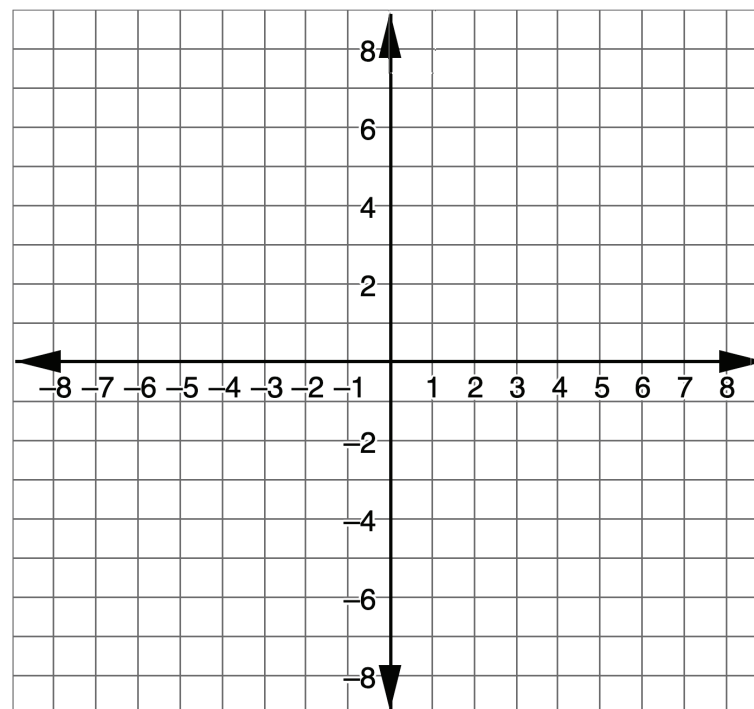
opposite sides of the x -axis, ☐ add
☐ subtract the absolute
value of the unlike coordinates. If the points are on the

same side of the x -axis, then ☐ add
☐ subtract the absolute
value of the unlike coordinates.



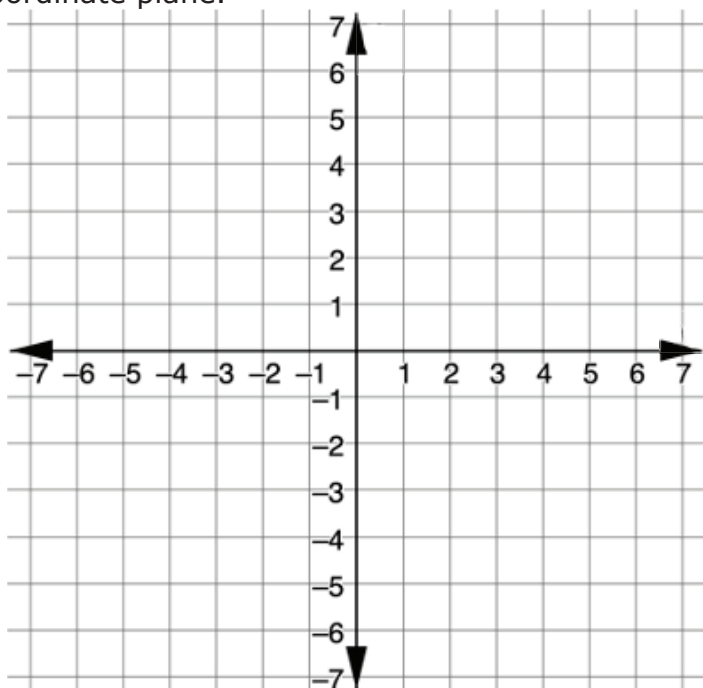
4.4.4 Guided Practice: Distances Between Rational Numbers

1. Graph the points $E(-5\frac{1}{3}, 4\frac{1}{2})$ and $F(7\frac{2}{3}, 4\frac{1}{2})$ on the coordinate plane.



- a. Describe the location of these points relative to the origin and the quadrants they lie in.
- b. What is the distance between point E and point F ?

2. Graph the points $K(-4, -3.25)$ and $M(-4, -0.5)$ on the coordinate plane.

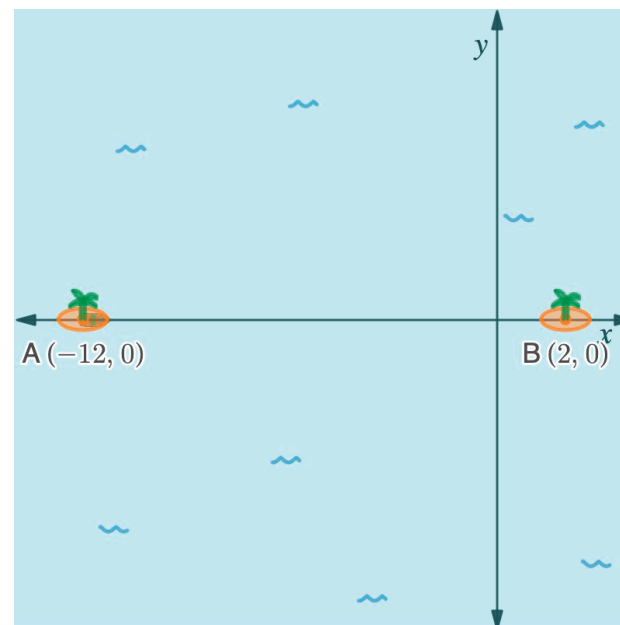


- a. Describe the location of these points relative to the origin and the quadrants they lie in.
- b. What is the distance between point K and point M ?

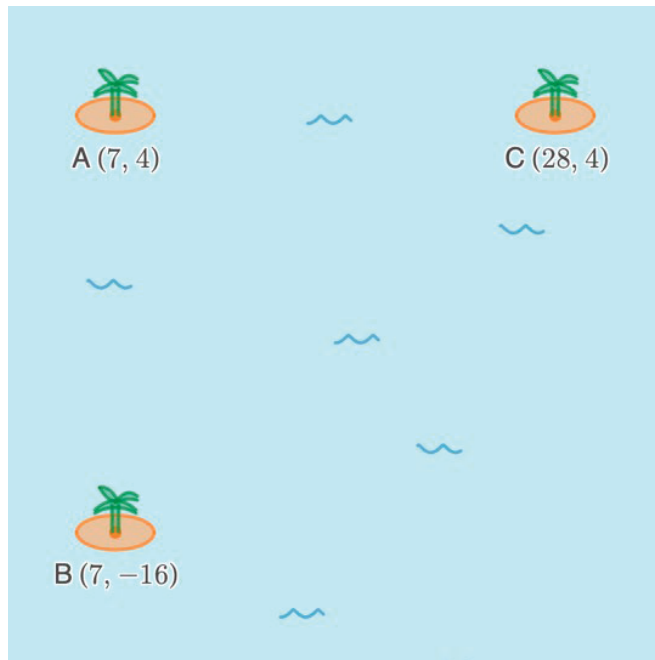


4.4.5 Your Turn!

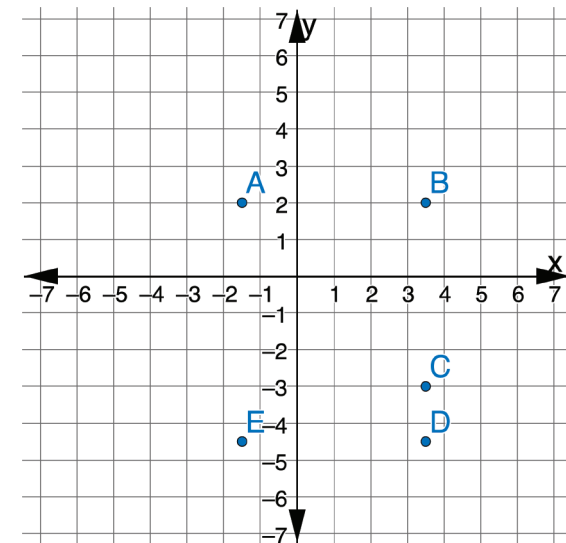
1. What is the distance from point A to point B ?



2. Explain which island is closest to island A . Show your work.



3. Five points are shown. Points A and E lie on the same vertical line. Points B, C , and D lie on a different vertical line. Points E and D lie on the same horizontal line.

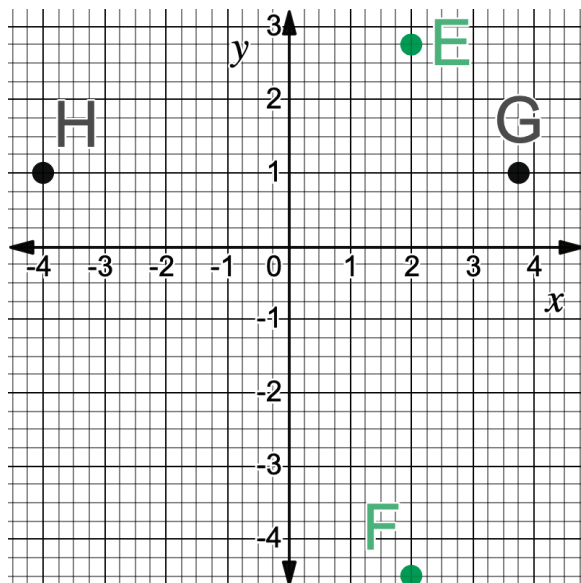


- a. Complete the table by filling in each missing coordinate.

A	$(-1.5, \underline{\quad})$
B	$(\underline{\quad}, 2)$
C	$(3.5, \underline{\quad})$
D	$(3.5, \underline{\quad})$
E	$(\underline{\quad}, -4.5)$

- b. The distance between points B and C is _____ units.
- c. The distance between points D and C is _____ units.
- d. The distance between points D and E is _____ units.

4. Use the coordinate plane below to determine distances between the following points.



- The distance between points E and F is _____ units.
- The distance between points G and H is _____ units.
- Which points are the same distance from $(2, 1)$?



4.4.6 Wrap-Up: Cool Down

- Your friend is absent today and will need the notes from class. Write a quick note to help them learn how to find the distance between two points on a coordinate plane.